When Context Cuts Both Ways: Divergent Effects of Contextual Cues on Semantic Processing in Monolinguals and Bilinguals

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A central question in the neurocognition of language is how the brain updates an evolving meaning representation with new semantic information, and how this process varies with language experience. The N400, a centro-parietal negativity linked to a temporal–parietal semantic network, indexes the ease of integrating a word with its prior context, showing reduced amplitude when context is richer [4, 5, 6]. Recent work quantifies this relationship using semantic distance, a distributional measure that predicts N400-like activity during naturalistic speech [1, 2]. However, the role of language experience in shaping these semantic-distance effects is less clear. ERP and neuroimaging studies suggest that bilinguals, especially in their non-dominant language (L2), often exhibit delayed or more broadly distributed N400 responses compared to monolinguals [6, 7, 3]. This raises the question of whether additional visual context facilitates or complicates semantic integration in L2 comprehension. Here, we use semantic distance as a computational probe to test how visual context modulates temporal–parietal semantic activity in monolingual and bilingual listeners during continuous speech.

We applied forward temporal response function (TRF) modeling to EEG recorded from 24 monolingual English speakers and 24 Mandarin–English bilinguals with mixed language dominance as they watched TED Talk excerpts in audio-only (AO) and audiovisual (AV) formats. Using the multivariate TRF toolbox, we modeled time-lagged EEG responses as a linear combination of semantic distance (i.e. GloVe cosine distance to the preceding 10-word context) and a few lexical covariates to capture evoked neural activity. TRF weights were estimated using ridge regression with cross-validation, followed by a current source density (CSD) transformation to improve spatial specificity and sharpen scalp topographies. We then examined these CSD-transformed TRFs in the 300–500 ms N400-range window.

Semantic distance produced robust N400-like effects that were differentially modulated by visual context in each group (Fig. 1). In AO, both monolinguals and bilinguals showed a clear negativity, consistent with an N400-like response in both groups. In AV, monolinguals showed a reliable reduction of this posterior negativity and an attenuated frontal positivity, consistent with previous findings that contextual cues ease integration demands. Bilinguals showed the opposite pattern: audiovisual speech significantly increased and spatially broadened the negativity relative to the AO condition. This suggests greater recruitment of semantic processing when additional visual cues are available. These effects were quantified using linear mixed-effects models on single-trial TRF amplitudes with fixed effects of Condition and 3D electrode coordinates; significant Condition × Topography interactions were observed in both groups (all $p < .001$).

These findings show that visual context shapes semantic integration in opposite ways depending on language experience. For monolinguals, visual context provides conceptual support that reduces the demands of semantic processing. For bilinguals, however, it introduces an information stream that may compete for attentional resources with non-dominant language, reflected by the larger scalp recorded negativities. Overall, our results suggested that while audiovisual context facilitates semantic processing in monolinguals, it may increase integration demands for bilinguals in their less dominant language.

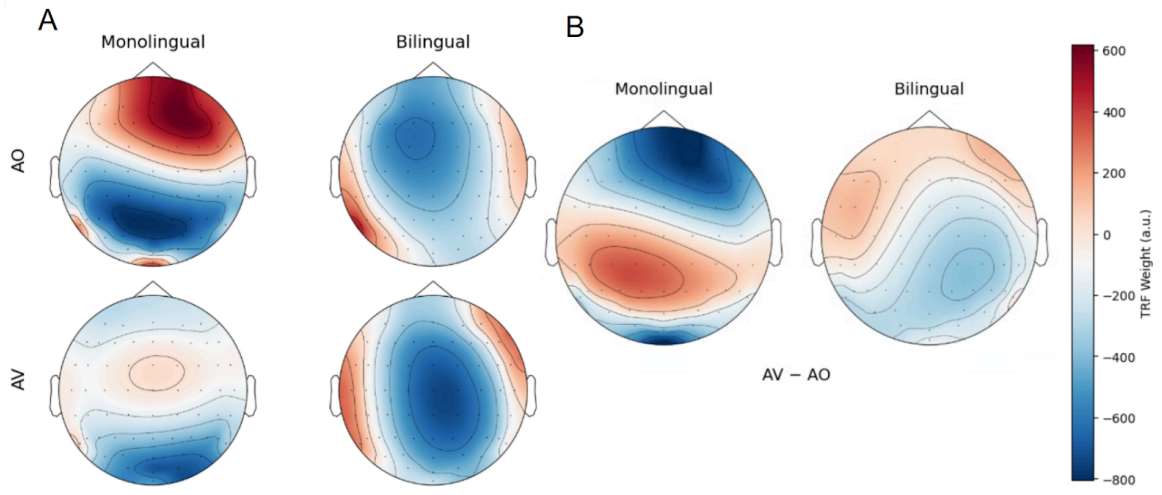

Figure 1. CSD-transformed TRF topographies for semantic distance in the 300–500 ms (N400) window. (A) TRF weights for monolingual and bilingual listeners in audio-only (AO) and audiovisual (AV) conditions. In AO, both groups show a N400-like negativity. Visual context modulates this pattern in opposite directions, where monolinguals show reduced posterior negativity in AV, and bilinguals show enhanced and more broadly distributed negativity. (B) AV–AO difference maps highlight this divergence, with monolinguals showing attenuated posterior negativity and bilinguals showing enhanced posterior negativity when visual context is available.

Reference:

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